United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam Trimester: Summer 2026
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (10 Marks) [CO3]

Global Descriptor Table	
Index	**Address**
0B4 H	Physical Address: 1D8E6A30, Limit: 3C7A2H, Access: 59H, G = 1
2C9 H	Physical Address: 63B2F094, Limit: E1D45H, Access: 96H, G = 0
1893 H	Physical Address: 48A3D6F8, Limit: 6B2F1H, Access: D3H, G = 0
756 H	Physical Address: 7F3C20B6, Limit: 25E8CH, Access: 2BH, G = 1
8E1 H	Physical Address: 0E94D7A2, Limit: B74C0H, Access: F1H, G = 0
1293 H	Physical Address: 5D61A8C4, Limit: 40B9DH, Access: C8H, G = 1

Local Descriptor Table	
Index	**Address**
03D H	Physical Address: 2A7C4E10, Limit: 4D1F3H, Access: 3AH, G = 0
1F8 H	Physical Address: 5B90E3A4, Limit: 0C2B7H, Access: F6H, G = 1
4B6 H	Physical Address: 71D4082C, Limit: 9A5E0H, Access: 8DH, G = 0
765 H	Physical Address: 4C3483B1, Limit: 1B3C6H, Access: B7H, G = 1
8C7 H	Physical Address: 36F5B7D9, Limit: 7E04AH, Access: 1CH, G = 1
B20 H	Physical Address: 90A1C5E2, Limit: 2F8D9H, Access: BAH, G = 0

The flight computer of the deep-space probe **Aurora-9** runs in protected mode on an imaginary 32-bit microprocessor called **Zephyr-32**, which uses the **same descriptor format as the Intel 80386**. The only difference is that the **G (Granularity) bit** in Zephyr-32 allows segment lengths to be specified in steps of **1 KB instead of 4 KB**. Part of its descriptor tables is shown above. A task requests a segment whose **index number is 1893 (Decimal)**, with a **Requested Privilege Level of 10 (Binary)**. Determine the following based on the above descriptor table:

a.	What is the value of the segment register? (Answer in hexadecimal.)	**2**
b.	For an offset value of **5E9A1 H**, determine the base address and the ending address of the segment.	**3**
c.	Has the segment been accessed?	**1**
d.	Segment type (**CS/DS/SS/ES**).	**2**
e.	Is the descriptor defined or undefined?	**1**
f.	What is the descriptor privilege level?	**1**

Question 2: Answer all the questions. (10 Marks) [CO5]

a.	An ADC system operates with a **15V reference voltage**. When an analog input voltage of **10V** is applied, the ADC reports a digital value of **10,922**. Determine **the number of bits** used by the ADC. **Show all necessary mathematical calculations.**	**3**
b.	In an **I^2C** communication protocol, there are a total of 15,000 slave devices. **How many bits are required to represent the slave addresses?** A microcontroller-based system has a master device: **"Master" (index: 60H)** and three slave devices: **"Slave 1" (index: 11H)**, **"Slave 2" (index: A9H)**, **"Slave 3" (index: B3H)**. Now, in **I^2C** data transfer protocol, Master receives **2-byte** data (from char "**FiNaL**") from **Slave 3**. Draw the corresponding sequence diagram. The ASCII value of 'G' is 71, and the ASCII value of 'g' is 103.	**1+3**
c.	A company is designing a processor for a system that requires high execution speed, efficient instruction pipelining, and a large number of registers. The processor should use simple, fixed-length instructions. The system can be built using either **RISC Architecture or CISC Architecture. Which architecture would be more suitable and why?**	**3**

Question 3: Answer all the questions. (10 Marks) **[CO3]**

a. Consider the following fetch cycle in an 8086 BIU: **3**

[Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch]

Assume that, during the execution of one instruction, the BIU fetches two instructions and saves them in the instruction queue. The 1st instruction is a '**MOV <address>**', the 4th instruction is a '**JUMP <7th instruction address>**', the 5th instruction is a '**JUMP <8th instruction address>**', the 7th instruction is a '**JUMP <5th instruction address>**' instruction. Draw the corresponding BIU and EU cycles.

b. The 8086 writes data to an I/O port. Draw the timing diagram for this operation showing the activities of the Address-Data and Address-Status pins ($AD_0 - AD_{15}$, $A_{16}/S_3 - A_{19}/S_6$), $\overline{ALE}$, DEN, CLK, WR, $DT/\overline{R}$, $\overline{M}/IO$, $\overline{RD}$ in each clock cycle. **[Observe the pins carefully]** **3**

Question 4: Answer all the questions. (10 Marks) **[CO4]**

RAM Content

Address	Content		Address	Content
8H	**0000 0111**		**CH**	**0011 1011**
9H	**0000 1001**		**DH**	**0000 0000**
AH	**0000 0000**		**EH**	**0000 0000**
BH	**0000 0000**		**FH**	**0000 0000**

a. Using the **RAM content table** given above, write an **assembly program** that shows the result of $7 \times 2^2 - 9$ on the **binary display**. **3**

b. **CON** = $C_P \overline{E_U}\,\overline{L_M}\,S_U \;\; L_A\,\overline{CE}\,\overline{L_I}\,E_I \;\; \overline{L_B}\,\overline{E_P}\,L_O\,E_A$ **[Observe the pins carefully]** **6**

At present, the **Accumulator** holds **2EH**, the **B Register** holds **1EH**, the **Instruction Register (IR)** holds **1FH**, the **Memory Address Register (MAR)** holds **6H**, and the **Output Register** holds **2DH**. Due to a fault, the controller-sequencer issues the following control words in **four consecutive T-states (T3 to T6)**:

i) **2CCH**, ii) **47CH**, iii) **624H**, iv) **364H**

Explain the **operation** carried out **in each T-state**. Also find the contents of the **Accumulator**, **B Register** and **Output Register** after each T-state, using the **RAM content table** given above.

[**Hint:** Only the **lower nibble** of the IR is sent to the MAR (e.g., IR = 7AH → MAR = AH).]

c. Using the **CON** given in **[b]**, write the **control words (in hexadecimal)** of the following instructions: **2+3**

i) **SUBOUT XH**: Subtracts the content of memory location **XH** from the Accumulator and shows the result on the **binary display**. The Accumulator remains unchanged. Write **T5 and T6** only (assume T4 is the same as the T4 of the **SUB** instruction).

[e.g., if Acc = 0AH and RAM[5H] = 03H, then **SUBOUT 5H** shows **07H** on the binary display (since, 0AH − 03H = 07H), and Acc stays **0AH**.]

ii) **OUTM XH**: Shows the content of memory location **XH** directly on the **binary display**. The Accumulator and B Register remain unchanged. Write **T4, T5 and T6**.

[e.g., if RAM[5H] = 3CH, then **OUTM 5H** shows **3CH** on the binary display.]